5.4.5 Capacitive Reactance

The resistance offered by a capacitor is called as capacitive reactance. The unit of capacitive reactance is ohm(Ω) and it is denoted by letters Xc.

$$Xc = \frac{1}{2\pi fc}$$

Where

Xc = capacitive reactance in ohm
C = capacitance in farad
F = frequency in Hertz

5.4.6 AC – RL SERIES CIRCUITS

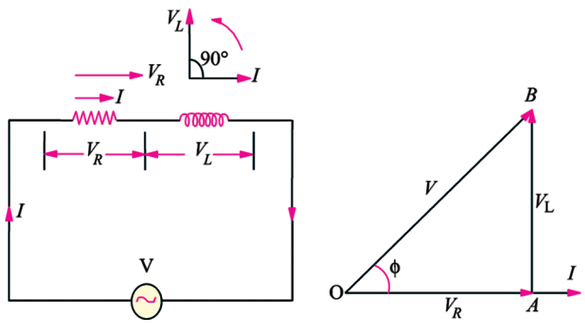


Fig 5.10 *AC through RL Series Circuit*

Resistance and inductance are connected in series as in fig. 5.10. In resistance there is no phase difference between Current and Voltage. But in inductance, current leads 90° by voltage.

V_R = Voltage across the resistance (Volts)

V_L = Voltage across the inductance (Volts)

R = Resistance (in ohms)

X_L = Inductive reactance

$V_R = I \times R$

$V_L = I \times X_L$

$V_2 = V_R^2 + V_L^2$

$V = \sqrt{V_R^2 + V_L^2}$

$V = \sqrt{(IR)^2 + (I X_L)^2}$

$V = \sqrt{I^2R^2 + I^2 X_L^2}$

$V = I\sqrt{R^2 + X_L^2}$

$\frac{V}{I} = \sqrt{R^2 + X_L^2}$

$Z = \sqrt{R^2 + X_L^2} \left(Z = \frac{V}{I} \right)$

5.4.7 AC-RC CIRCUITS

In fig. 5.11 resistance and capacitance are connected in series. In resistance there is no phase difference between current and voltage. But in capacitance, voltage is 90° lagging by current.

V_R = Voltage across the resistance (Volts)

V_L = Voltage across the inductance (Volts)

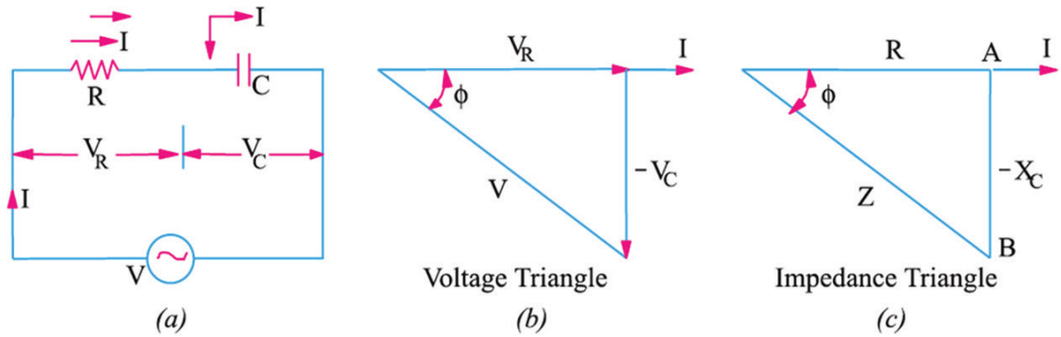


Fig 5.11 *AC through RC Series Circuit*

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